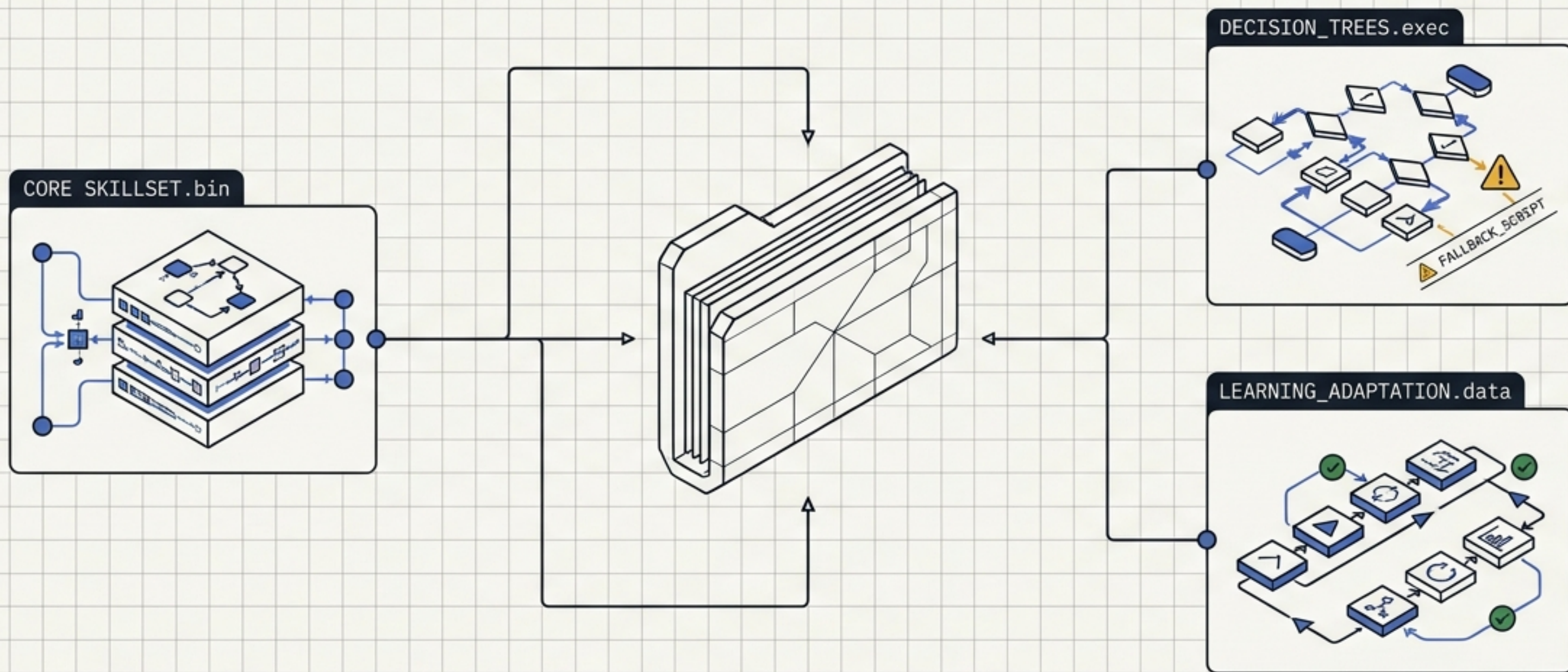


Procedural Memory for AI



The Architecture and Ecosystem of Agent Skills

AI Agents Know Facts. They Lack Procedure.

Knowledge_Query.sh

> What is the airspeed velocity of an unladen swallow?



The airspeed velocity of an unladen European swallow is roughly 11 meters per second, or 24 miles per hour. An African swallow is non-migratory.

Workflow_Task.sh

> Generate the 47-step compliant financial report.



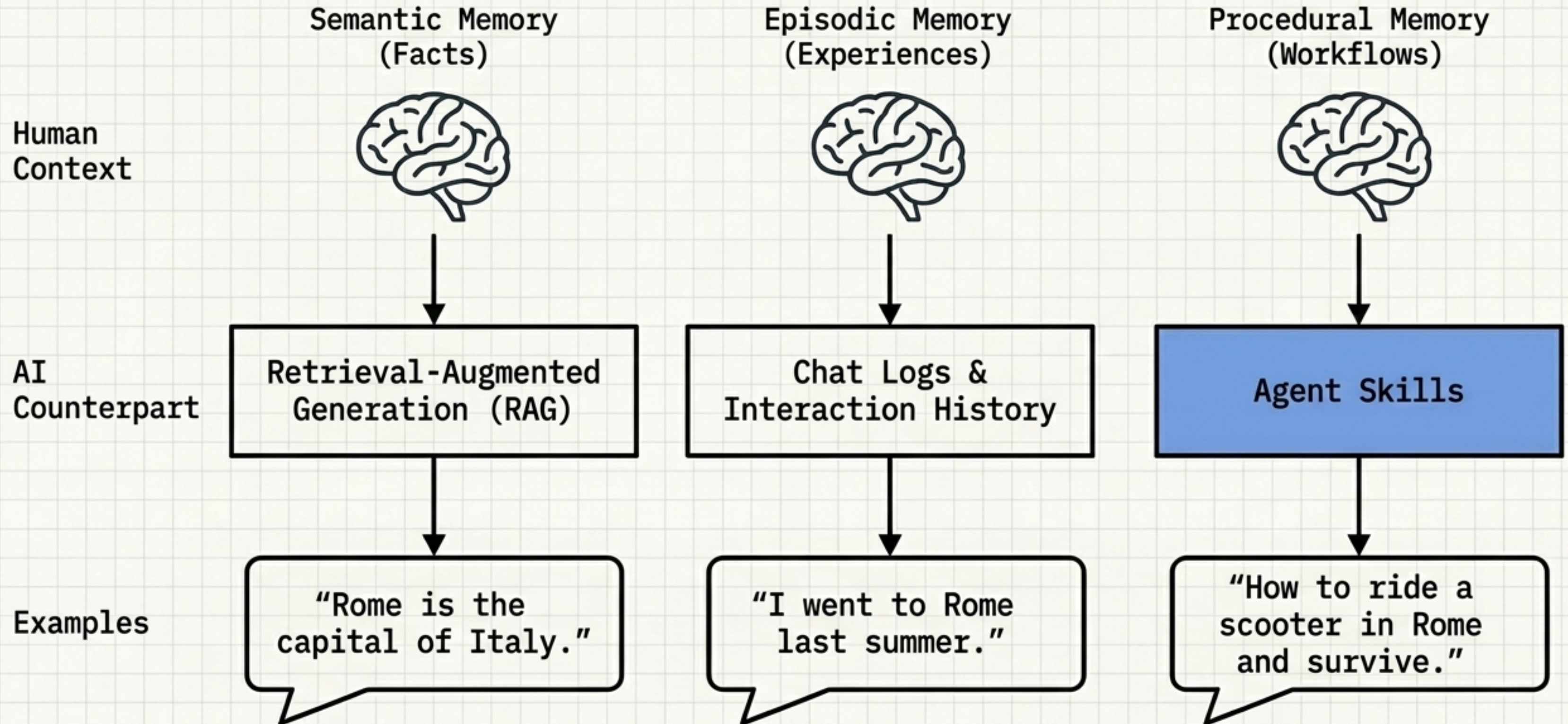
```
STEP 01: INITIATE SESSION...  
[ERR] UNDEFINED PROCEDURE FOR COMPLIANCE MODULE...  
>>> print(f'Generating report step {STEP_NUM}')
```

// Skipped steps 2-14...

```
MISSING: 'DATA_VALIDATION_KEY'...  
[WARNING] HALLUCINATED REGULATORY REFERENCE:  
'Section 404b.7C (2023)'...  
END GENERATION...  
FAILED TO COMPLETE.
```

Large Language Models are excellent reasoners and encyclopedias.
But without explicit, step-by-step guidance, they are forced to
guess at complex, company-specific workflows.

The Cognitive Architecture of Artificial Intelligence



Defining the AI Knowledge Stack

Paradigm	Memory Type	Primary Function	Token Overhead	Update Difficulty
Model Context Protocol (MCP)	Tool Access	External API Reach	Low	Moderate
RAG	Facts	Reference Lookup	High	Low
Fine-Tuning	Weights	Tone/Innate Knowledge	None	High (Expensive)
Agent Skills	Procedure	Workflow Execution	Progressive (Efficient)	Low (Markdown/Git)

MCP provides the reach to invoke external APIs.
Agent Skills provide the judgment for when and how to do it.

What is an Agent Skill?

A **standardized, open format** for extending AI agent capabilities with specialized knowledge and workflows. At its core, a skill is a portable, **version-controlled folder** containing a SKILL.md file.



Domain Expertise

Captures specialized knowledge (legal review, data analysis).

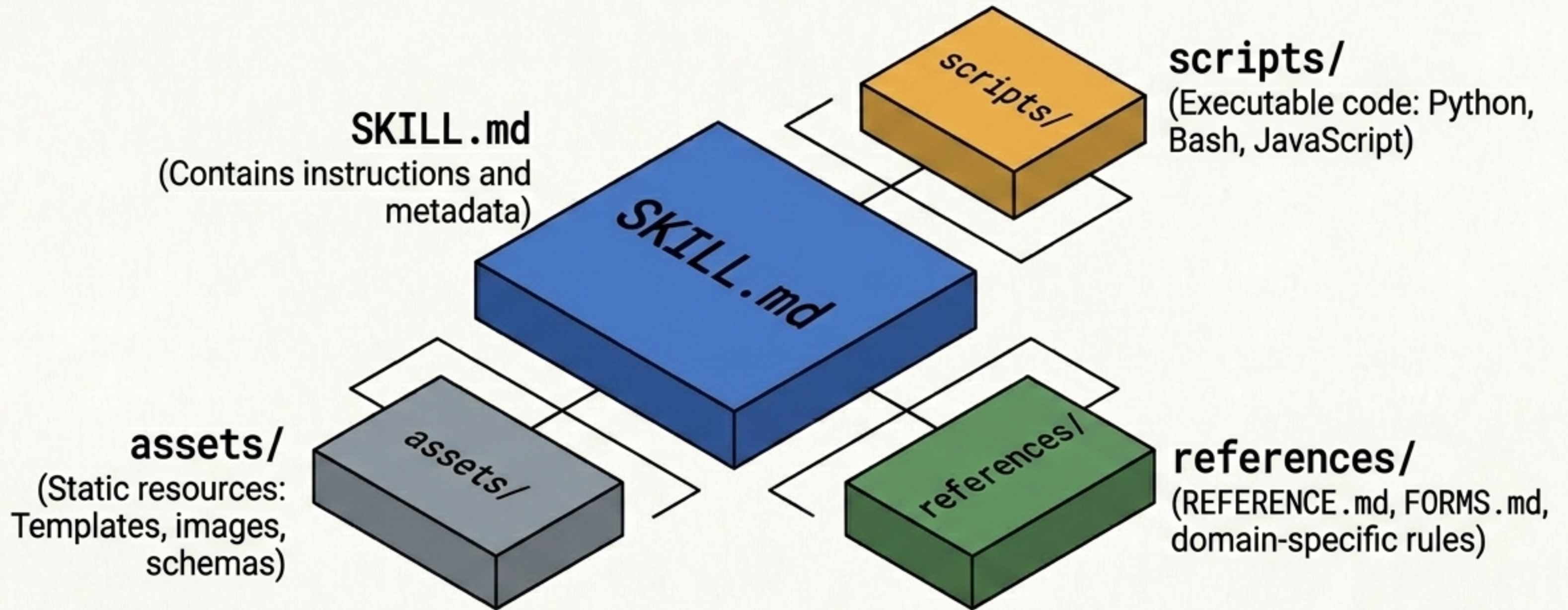
Repeatable Workflows

Turns multi-step tasks into auditable procedures.

Cross-Product Reuse

Build once, deploy across any skills-compatible agent.

Anatomy of an Agent Skill



Keep the main SKILL.md under 500 lines. Offload execution to scripts and deep context to references.

The Trigger Condition: YAML Frontmatter

name: pdf-builder

description: Use this when the user asks to extract data from a PDF.

license: Apache-2.0

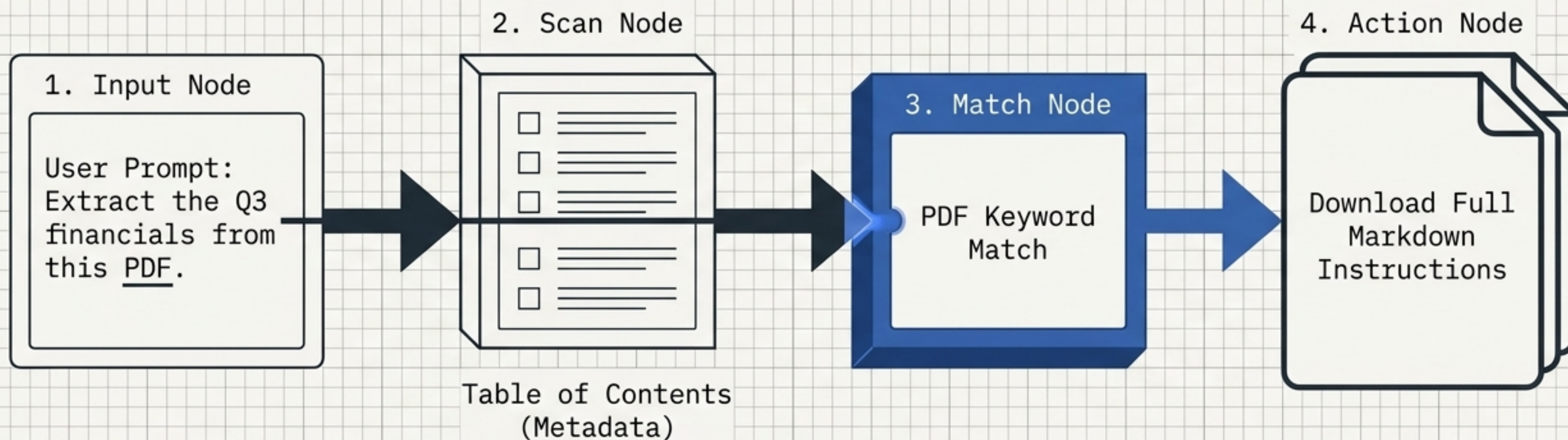
The Identity.

Max 64 characters, lowercase, alphanumeric and hyphens only. Matches the parent directory name.

The Trigger Condition.

Max 1024 characters. Crucially, this teaches the LLM when to activate the skill through keyword matching and reasoning.

How Agents Route Workflows

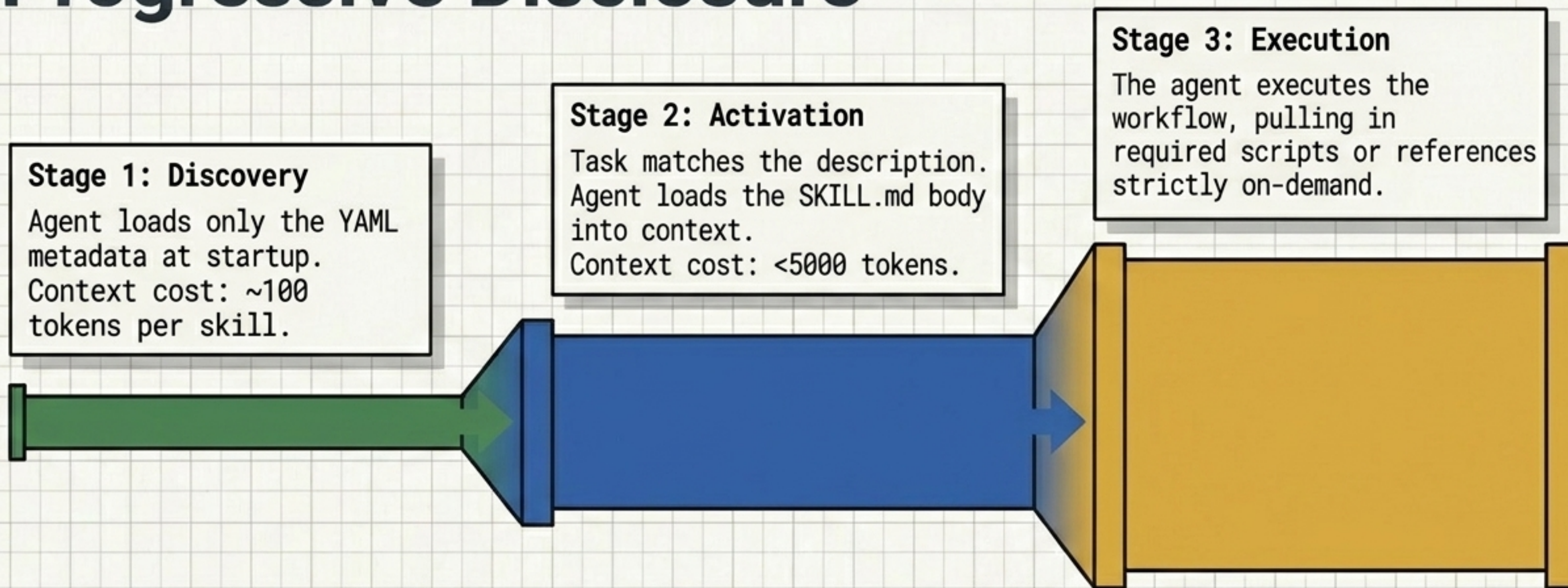


The Token Budget Problem

If an agent has hundreds of installed skills, loading all full instructions and scripts at startup would **blow through the** entire token budget before the user even asks a question.

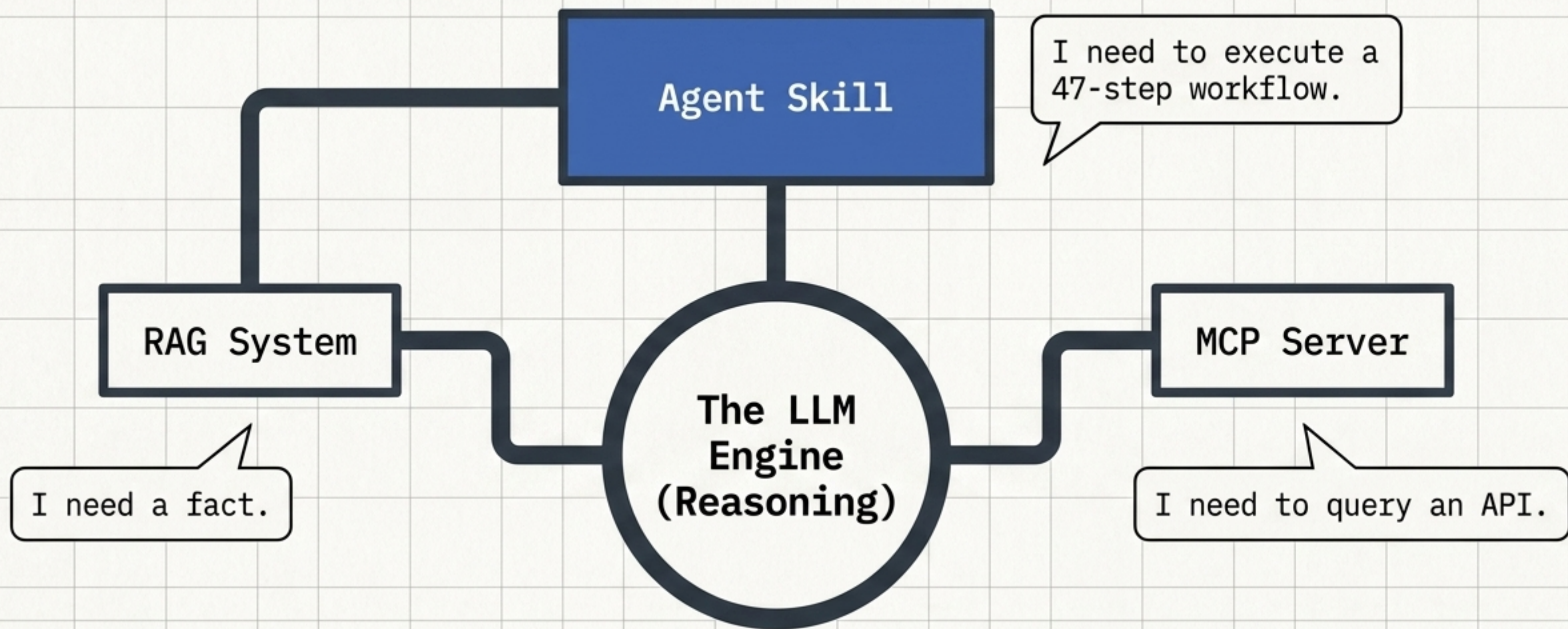


Progressive Disclosure



Full instructions load only when a task calls for them, keeping the resting context footprint **incredibly small**.

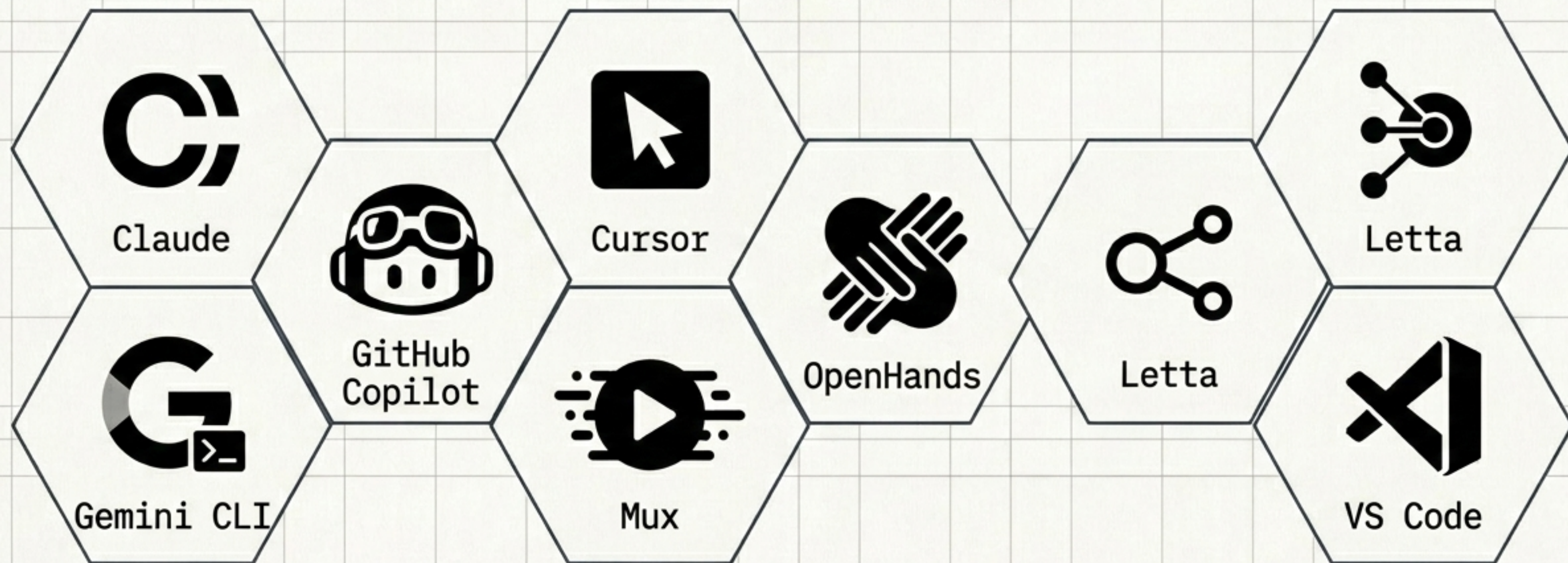
The Complete Agent Architecture



Agent Skills act as the orchestrator. They are the procedural judgment layer that tells the agent when to pull a fact from RAG, and when to pull a lever via MCP.

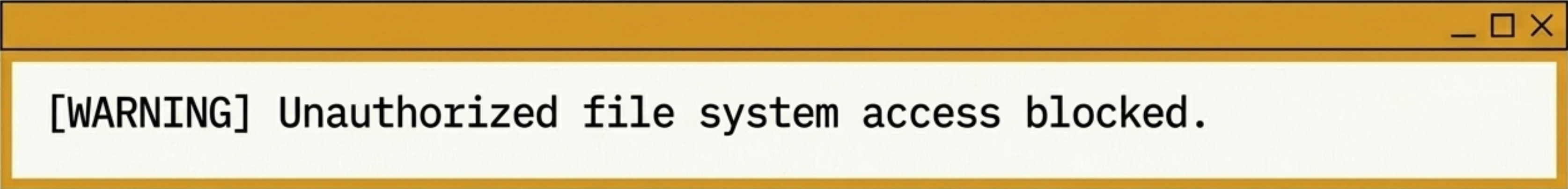
An Open Standard Ecosystem

Originally developed by Anthropic and released as an Apache 2.0 open standard. A skill built for one platform works on any platform that supports the specification. Write once, run anywhere.



Execution Power Requires Execution Trust

Skills can include executable scripts with access to local file systems, environment variables, and API keys.

A yellow rectangular warning box with a black border and standard window control buttons (minimize, maximize, close) in the top right corner.

[WARNING] Unauthorized file system access blocked.

Prompt injection

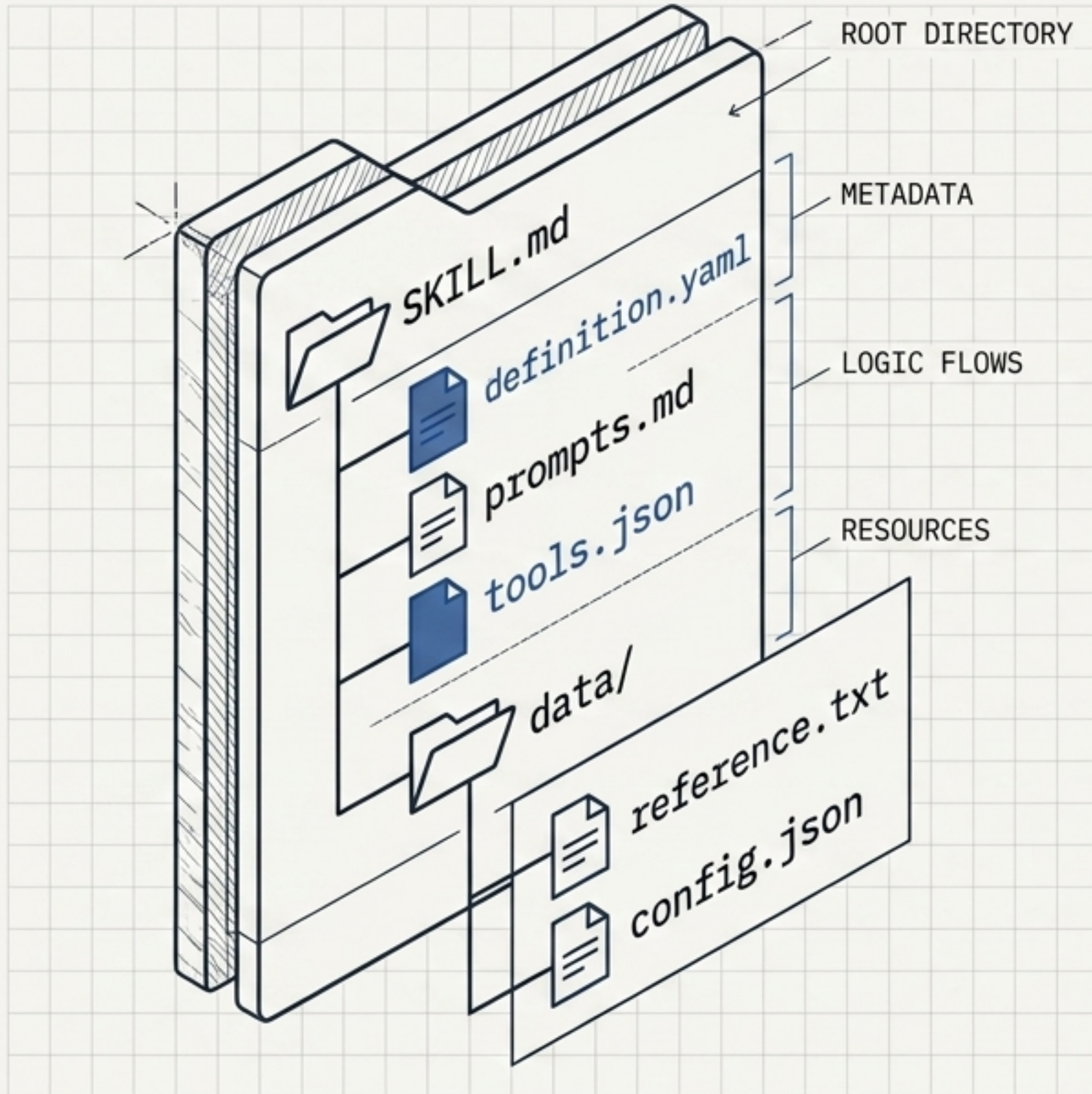
Tool poisoning

Hidden malware

A square icon with diagonal black and yellow stripes, commonly used to denote a warning or hazard.

The Golden Rule: Treat skill installation exactly like installing any external software dependency. Review the code before running it locally.

The Procedural Era of AI



1. Format

Defined in a lightweight, open-standard Markdown file and directory.

2. Efficiency

Conditionally triggered and loaded via token-efficient progressive disclosure.

3. Impact

Agents that already know the airspeed velocity of a swallow can now be taught to perform your exact, repeatable workflows.